Classification of quantum graphs over complex 3×3 matrices

(Klasyfikacja grafów kwantowych nad
zespolonymi macierzami 3×3)

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Master's thesis

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Abstract

According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible.

Według wszystkich znanych praw lotnictwa, pszczoła nie powinna w ogóle latać. Jej skrzydła są zbyt małe, by unieść jej pulchne ciałko z ziemi. Pszczoła, oczywiście, lata mimo wszystko, ponieważ pszczoły nie przejmują się tym, co ludzie uważają za niemożliwe.

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Chapter 1

Introduction

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1.3 Group theory

In the following section we will introduce some group theoretic notions that be used throughout this thesis. We assume that the Reader is familiar with the notion of a group.

Definition 1.3.1 (Dual of a group)**.** Let G be a group. The *dual group* $\hat{G}$ is the set of all homomorphisms from G into the multiplicative group of roots of unity in $\mathbb{C}$ with the group operation defined as

$$(\varphi\psi)(g) = \varphi(g)\psi(g)$$

for all $\varphi, \psi \in \hat{G}$ and $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

Remark 1.3.1. We will often call the elements of the dual group *characters*. If a group is isomorphic to its complement it will be called *self-dual*.

The following proposition will be used throughout the thesis, as almost all of the groups we will be dealing with will be finite and abelian.

Proposition 1.3.1. *If G is finite and abelian, then it is self-dual.*

Proof. First we will prove the statement for finite cyclic groups. Let $G = \langle g \rangle$, $\operatorname{ord}(g) = n$. We will prove that the dual is cyclic of order n. Let $\chi : G \to \mathbb{C}$ be a character. Since the group is cyclic, the character is entirely determined by the value it takes on the generator g. Since $g^n = e$, obviously $\chi(g)^n = 1$, so $\chi(g)^n \in \mu_n = \{\exp(2k\pi i/n) : k = 0, 1, \ldots, n-1\}$. For each possible choice of $\chi(g) \in \mu_n$

we get a different character. It is then easy to see that $\chi : G \to \mathbb{C}$ such that $\chi(g) = \exp(2\pi i/n)$ generates this group, i.e. $\hat{G} = \{\chi, \chi^2, \ldots, \chi^{n-1}, \chi^n = e_{\hat{G}}\}$.

Now since every finite abelian group can be written as a finite product of finite cycles, all we need to do is check whether the statement holds for a product of two groups A, B that are themselves self-dual, i.e. $\widehat{A \times B} \cong \widehat{A} \times \widehat{B}$. This is self-evident as every character $\chi \in \widehat{A \times B}$ is determined by its restrictions to $A \times \{e_B\}$ and $\{e_A\} \times B$, since

$$\chi(a, b) = \chi((a, e_B)(e_A, b)) = \chi(a, e_B)\chi(e_A, b).$$

Thus $\chi \leftrightarrow (\chi|_A, \chi|_B)$ is an isomorphism. $\square$

Definition 1.3.2 (Group action on a C^*-algebra)**.** By *group action* of a group G on a C^*-algebra A we mean a group homomorphism

$$\alpha : G \to \mathrm{Aut}(A).$$

We will write $g \cdot a$ instead of $\alpha(g)(a)$ then the action in question is clear.

1.4 C^*-algebras

In the following section we will introduce some notions stemming from the theory of C^*-algebras. We assume the Reader is familiar with the notion of a C^*-algebra. For reference, see Blackadar [1].

Multiplication in a C^*-algebra A is a bilinear map

$$m\colon A \times A \to A.$$

By the universal property of the tensor product, m extends uniquely to a linear map

$$m\colon A \otimes A \to A.$$

Throughout this thesis, we will use the same notation (m) for both the bilinear multiplication and its unique linear extension, relying on the context to distinguish between the two.

Definition 1.4.1 (Linear functional)**.** A *linear functional* on a C^*-algebra A is a linear function $\psi : A \to \mathbb{C}$.

Remark 1.4.1. We will call linear functionals simply as *functionals*.

The following notions will be useful for defining an inner product on C^*-algebras. Positiveness and faithfulness are all we need to define a norm on a finite-dimensional C^*-algebra equipped with a suitable functional.

Definition 1.4.2 (Positive functional)**.** A functional ψ on a C^*-algebra A is called *positive*, if

$$\psi(a^*a) \geq 0$$

for any $a \in A$.

Definition 1.4.3 (Faithful functional)**.** A positive functional ψ on a C^*-algebra A is called *faithful*, if

$$\psi(a^*a) = 0 \Leftrightarrow a = 0$$

for any $a \in A$.

Remark 1.4.2. In the literature, the functional ψ is often called a *state*.

1.4.1 Crossed-product C^*-algebra

In this section we will introduce the notion of a crossed-product C^*-algebra. We will do so in a limited manner, only to the extent to which the theory of such products is necessary for our purposes. For a more general reference we refer the Reader to Blackadar [1].

The following definitions are based off of Sierakowski [5].

Definition 1.4.4 (C^*-dynamical system)**.** A C^*-dynamical system is a triple (A, G, α), where A is a C^*-algebra, G is a group, and $\alpha : G \to \mathrm{Aut}(A)$ is a group action on A. For a C^* dynamical system (A, G, α) with G finite we define

$$C_c(G, A, \alpha) = \{f : G \to A : \mathrm{supp}(f) \text{ is finite}\},$$

i.e. the space of all finitely supported functions on G with values in A.

$C_c(G, A, \alpha)$ is equipped with both a product and a $*$-operation in such a way that the action becomes "inner", i.e. $g \cdot a = u_g a u_g^*$ for all $g \in G$, $a \in A$. More precisely we define those two operations in the following way:

$$ab = \sum_{g,h \in G}$$

Definition 1.4.5 (Covariant representation)**.** Fix a dynamical system (A, G, α). A *covariant representation* (π, u, H) of (A, G, α) consist of a unitary representation $u : G \to B(\mathcal{H})$ of G and a representation $\pi : A \to B(\mathcal{H})$ of A such that $u(g)\pi(a)u(g)^* = \pi(g \cdot a)$ for every $g \in G$ and $a \in A$.

Definition 1.4.6 (Full C^*-algebra norm on $C_c(G, A, \alpha)$)**.** The full C^*-algebra norm on $C_c(G, A, \alpha)$ is given by

$$\|\cdot\| = \sup \|(\pi \times u)(\cdot)\|,$$

where the supremum is taken over all covariant representations (π, u, H) of (A, G, α).

Definition 1.4.7 (Crossed product C^*-algebra)**.** The crossed product of a C^*-algebra and a discrete group G is the completion of $C_c(G, A, \alpha)$ with respect to the full norm. We denote the crossed product by $\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}$.

1.5 Quantum sets

Let B denote a C^*-algebra and let ψ denote a functional on this algebra. By the Gelfand-Naimark theorem, every C^*-algebra is isometrically $*$-isomorphic to a C^*-subalgebra $B(\mathcal{H})$ for some Hilbert space $\mathcal{H}$. If B is finite-dimensional, one can identify $L^2(B)$ with B and think of B as a Hilbert space with an inner product given by

$$\langle a, b\rangle_\psi = \psi(a^* b).$$

The following definition introduces one of the central notions used in this thesis. It is taken directly from Schäfer i Tobolski [4].

Definition 1.5.1 (Quantum set, Schäfer i Tobolski [4], Definition 2.1)**.** A quantum set (B, ψ) is a pair consisting of a finite-dimensional C^*-algebra B and a faithful positive functional $\psi : B \to \mathbb{C}$ such that

$$mm^* = \mathrm{Id}_B$$

holds for the multiplication map $m : B \otimes B \to B$ and its adjoint m^* given by

$$\langle m(a \otimes b), c\rangle = \langle a \otimes b, m^*(c)\rangle_{\psi\otimes\psi}$$

for all $a, b, c \in B$.

Remark 1.5.1. We will call any functional with the property given in the definition above a *quantum set functional* on B.

Remark 1.5.2. Recall that in general, a *comultiplication* on a C^*-algebra is a $*$-homomorphism

$$\Delta : A \to A \otimes A$$

such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & A \otimes A \arrow[d, "\Delta\otimes\mathrm{id}"] \\ A \otimes A \arrow[r, "\mathrm{id}\otimes\Delta"'] & A \otimes A \otimes A \end{tikzcd}$$

Note that in general a comultiplication is *not* unique. In our setting it is uniquely determined by the additional structure and the Riesz representation theorem for Hilbert spaces.

Remark 1.5.3. Since in our setup the comultiplication is given by the scalar product and regular multiplication m, we will denote it by m^*.

Example 1.5.1. Every finite set X gives rise to a quantum set $(C(X), \psi_X)$ with $\psi_X : C(X) \to \mathbb{C}$, $\psi_X(f) = \sum_{x\in X} f(x)$ for any $f \in C(X)$. Indeed, with this setup $m^*(e_x) = e_x \otimes e_x$ holds:

Example 1.5.2. In particular, $(\mathbb{C}^n, \psi_n)$ with $\psi_n(e_i) = 1$ is a quantum set. In this case...

Example 1.5.3. For any $n \in \mathbb{N}_+$, $(M_n, n\,\mathrm{Tr})$ is a quantum set, with Tr being the trace.

Lemma 1.5.1 (Comultiplication for matrix algebras)**.** *In the context of* $(M_n, n\,\mathrm{Tr})$*, the comultiplication is given by the formula*

$$m^*(e_{ij}) = \sum_{k=1}^{n} e_{ik} \otimes e_{kj}$$

with e_{ij} *being the matrix with a single* 1 *in the* i*-th row and* j*-th column, and extended linearly.*

Proof. This is simply a computational excercise. We will use the following well-known formulas:

1. $e_{ab}e_{cd} = \delta_{bc}e_{ad}$,
2. $\langle e_{ij}, e_{kl}\rangle = \mathrm{Tr}(e_{ji}e_{kl}) = \delta_{ik}e_{jl}$,

with

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$$

In particular $\{e_{ij}\}_{i,j=1,\dots,n}$ form a orthonormal basis.

Take arbitrary basis vectors e_{ab}, e_{cd}. Then

$$m(e_{ab} \otimes e_{cd}) = e_{ab}e_{cd} = \delta_{bc}e_{ad},$$

hence

$$\begin{aligned} \langle m(e_{ab} \otimes e_{cd}), e_{ij}\rangle &= \delta_{bc}\langle e_{ad}, e_{ij}\rangle \\ &= \delta_{bc}\delta_{ai}\delta_{dj}. \end{aligned}$$

Now compute the inner product with the proposed adjoint:

$$\begin{aligned} \left\langle e_{ab} \otimes e_{cd}, \sum_{k=1}^{n} e_{ik} \otimes e_{kj} \right\rangle &= \sum_{k=1}^{n} \langle e_{ab}, e_{ik}\rangle \langle e_{cd}, e_{kj}\rangle \\ &= \sum_{k=1}^{n} \delta_{ai}\delta_{bk}\delta_{ck}\delta_{dj} \\ &= \delta_{ai}\delta_{bc}\delta_{dj}. \end{aligned}$$

The two expressions agree. Since the matrix unix form an orthonormal basis, this linearly extends to all matrices. $\square$

1.5.1 Crossed-product quantum sets

Schäfer i Tobolski [4] have introduced the notion of crossed product quantum sets. Suppose we are given a finite-dimensional C^*-algebra $\tilde{B}$ along with a dual group action $\hat{a} : \hat{G} \to \mathrm{Aut}(\tilde{B})$ where G is finite and abelian. One can then consider the crossed product

$$B := \tilde{B} \rtimes_{\hat{\alpha}} \hat{G}.$$

If $\tilde{B}$ is equipped with a faithful functional $\tilde{\psi}$ in such a way as to make $(\tilde{B}, \tilde{\psi})$ a quantum set, one can define a new faithful functional on B to make it a quantum set, as described below.

Definition 1.5.2 (Crossed product quantum set, Schäfer i Tobolski [4], Definition 3.1)**.** Let $(\tilde{B}, \tilde{\psi})$ be a quantum set and let $\hat{\alpha} : \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$ be an action of the dual of a finite abelian group G. The *crossed product quantum set* is defined as

$$(\tilde{B}, \tilde{\psi}) \rtimes_{\hat{\alpha}} \hat{G} := \left(\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}, \psi_{\rtimes}\right),$$

where $\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}$ is the crossed product C^*-algebra and $\psi_{\rtimes}$ is the functional given by

$$\psi_{\rtimes}\left(\sum_{\chi \in \hat{G}} b_\chi u_\chi\right) = n\tilde{\psi}(b_1)$$

with $n = |\hat{G}|$.

The proof that this definition is sensible, i.e. the newly defined object is indeed a quantum set, can be seen in Schäfer i Tobolski [4], Proposition 3.4.

1.6 Quantum graphs

A simple graph is classically defined as a pair (V, E) with V serving as the *set of vertices* and $E \subseteq [V]^2$ serving as the *set of edges*. Such a graph could be interpreted as a matrix $A = [a_{ij}] \in \{0, 1\}^{n \times n}$ with $\{v_i, v_j\} \in E \Leftrightarrow a_{ij} = 1$; such a representation is often called the *adjacency matrix*. The key property of such matrices is that they are invariant with respect to entrywise multiplication. In turn we can use this property to generalise graphs on sets to graphs on quantum sets.

The following definitions are due to Schäfer i Tobolski [4].

Definition 1.6.1 (Quantum adjacency matrix, Schäfer i Tobolski [4], Definition 2.5)**.** Let (B, ψ) be a quantum set. A *quantum adjacency matrix* on this quantum set is a linear map $A : B \to B$ which is Schur-idempotent and *-preserving, by which we mean that it satisfies

$$m(A \otimes A)m^* = A, \qquad A(b^*) = A(b)^* \quad \text{for all } b \in B.$$

Definition 1.6.2 (Quantum graph, Schäfer i Tobolski [4], Definition 2.5)**.** A *quantum graph* on (B, ψ) is given by a quantum adjacency matrix on this quantum set.

For all intents and purposes we will identify both notions and simply call an appropriate $A : B \to B$ both a quantum adjancenty matrix and a quantum graph.

Remark 1.6.1. There are other equivalent definitions of a quantum graph, namely...

Chapter 2

Quantum graphs over 3×3 matrices

The classification on 2×2 matrices was done by Matsuda [3].

2.1 Gromada's theorem

In the following section we will state...

The following theorem is due to Gromada [2].

Theorem 2.1.1 (Gromada [2], Theorem 3.7). *Consider $m \in \{0, 1, \ldots, n^2\}$. Quantum graphs on M_n with m quantum edges are in a one-to-one correspondence with m-dimensional subspaces $V \subseteq \mathfrak{gl}_n$. The adjacency matrix is given by $A_V = \sum_{s=1}^{m} \xi_s \otimes \xi_s^*$, where $(\xi_1, \xi_2, \ldots, \xi_m)$ is some orthonormal basis of V.*

The quantum graph has no loops if and only if $V \subseteq \mathfrak{sl}_n$. It is undirected if and only if $V = V^*$.

Proof. See Gromada's paper for reference. □

Lemma 2.1.1 (Gromada [2], Lemma 3.8). *Any vector subspace $V \subseteq \mathfrak{gl}_n$ with $V = V^*$ has a self-adjoint basis $\{\xi_s\}$, $\xi_s = \xi_s^*$.*

Proof. We construct the basis by induction. Suppose we already have a basis $\{\xi_s\}$ of a subspace $V_1 \subseteq V$ such that $\xi_s = \xi_s^*$. This means that $V_1 = V_1^*$. Let V_2 be complement to V_1. Take any $\xi \in V_2$ and consider $(\xi + \xi^*)/2$ and $i(\xi^* - \xi)/2$. Both elements are self-adjoint and elements of V_2. They might or might not be linearly independent, but definitely at least one of them is non-zero. So, include both (if they are linearly independent) or one of them (if they are not) to our basis $\{\xi_s\}$ and repeat. □

Remark 2.1.1. The following lemma tells us that any undirected graph on M_n can be thought of as being made from symmetric quantum edges only.

Remark 2.1.2. A complex vector space $V = V^* \subseteq \mathfrak{gl}_n$ uniquely determines a real vector space spanned by the basis $\{\xi_s\}$ of self-adjoint elements $\xi_s = \xi_s^*$. Indeed, it is easy to see that any other self-adjoint element $\xi = \xi^*$ must be a real linear combination of $\{\xi_s\}$.

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